

Concept and Purpose

The purpose of our Shiny app is to provide an interactive and visually engaging representation of measles cases reported in the United States since January 1, 2025. Although measles was declared eliminated in the United States in 2000 by the CDC, recent years have shown a concerning resurgence in reported cases. While current case counts remain far below the levels observed before the widespread use of the measles vaccine, the nation is experiencing the highest number of measles cases seen since the early 1990s. This trend highlights the continued public health importance of monitoring outbreaks and understanding patterns of transmission. Our app is designed to make complex surveillance data easier to interpret by transforming raw case counts into accessible maps, charts, and summaries. By allowing users to explore measles activity geographically and over time, the app aims to improve awareness of outbreak trends, support public health education, and provide a centralized tool for examining the reemergence of a vaccine-preventable disease in the United States.

Initial Plan / Motivation

Our initial plan for the Shiny app was to create a comprehensive dashboard that allowed users to analyze measles data at both the county and state levels. We intended for the app to display case counts stratified by multiple key factors, including whether infections were imported or locally acquired, as well as whether affected individuals were vaccinated or unvaccinated. Organizing the data based on these parameters would provide a robust and detailed understanding of how outbreaks are occurring across the country. To enhance usability, we planned separate tabs for county-level and state-level summary statistics so users could easily compare regions and identify areas with elevated case counts. In addition, we proposed a correlation analysis section to explore relationships between important outbreak variables. Specifically, this portion of the app was motivated by questions such as whether increases in unvaccinated cases are associated with greater local transmission, and whether higher numbers of imported cases contribute to additional local spread. These analytical features were intended to move the app beyond simple visualization and toward meaningful epidemiological insight.

Initial concepts for Shiny app layouts are shown below:

This layout has tabs and tables and can be viewed [here](#).

U.S. Measles Case Explorer

Filters

Explore measles cases by location and time.

Select State:

All

Select County:

All

Select Date Range:

2025-01-16

to

2026-04-02

☒ Show points on graph

Overview

Trends

Comparison

Total Reported Cases

3981

Affected Locations

301

Date Range

2025-01-16 to 2026-04-02

Filtered Data Preview

Show 10 entries

Search:

	location_name	location_id	location_type	date	outcome_type	value	state	county
1	Adams, Colorado	8001	county	2025-08-13	case_lab-confirmed	1	Colorado	Adams
2	Adams, Colorado	8001	county	2025-08-15	case_lab-confirmed	2	Colorado	Adams
3	Adams, Colorado	8001	county	2025-08-28	case_lab-confirmed	1	Colorado	Adams
4	Adams, Colorado	8001	county	2026-03-06	case_lab-confirmed	5	Colorado	Adams
5	Alachua, Florida	12001	county	2026-01-25	case_lab-confirmed	1	Florida	Alachua
6	Alachua, Florida	12001	county	2026-02-07	case_lab-confirmed	1	Florida	Alachua
7	Allegan, Michigan	26005	county	2025-06-06	case_lab-confirmed	1	Michigan	Allegan
8	Allen, Indiana	18003	county	2025-04-07	case_lab-confirmed	1	Indiana	Allen
9	Allen, Indiana	18003	county	2025-04-09	case_lab-confirmed	5	Indiana	Allen
10	Allen, Indiana	18003	county	2025-04-23	case_lab-confirmed	2	Indiana	Allen

Showing 1 to 10 of 1,058 entries

Previous

1

2

3

4

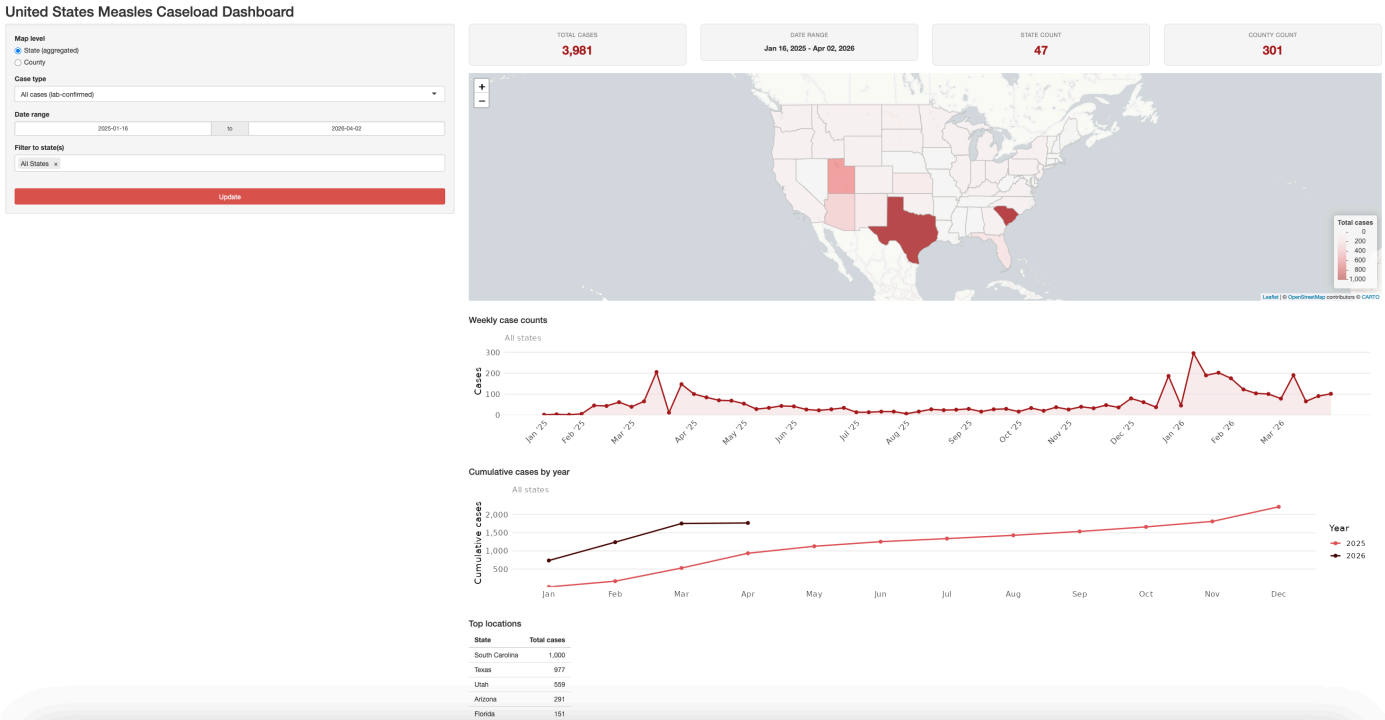
5

...

106

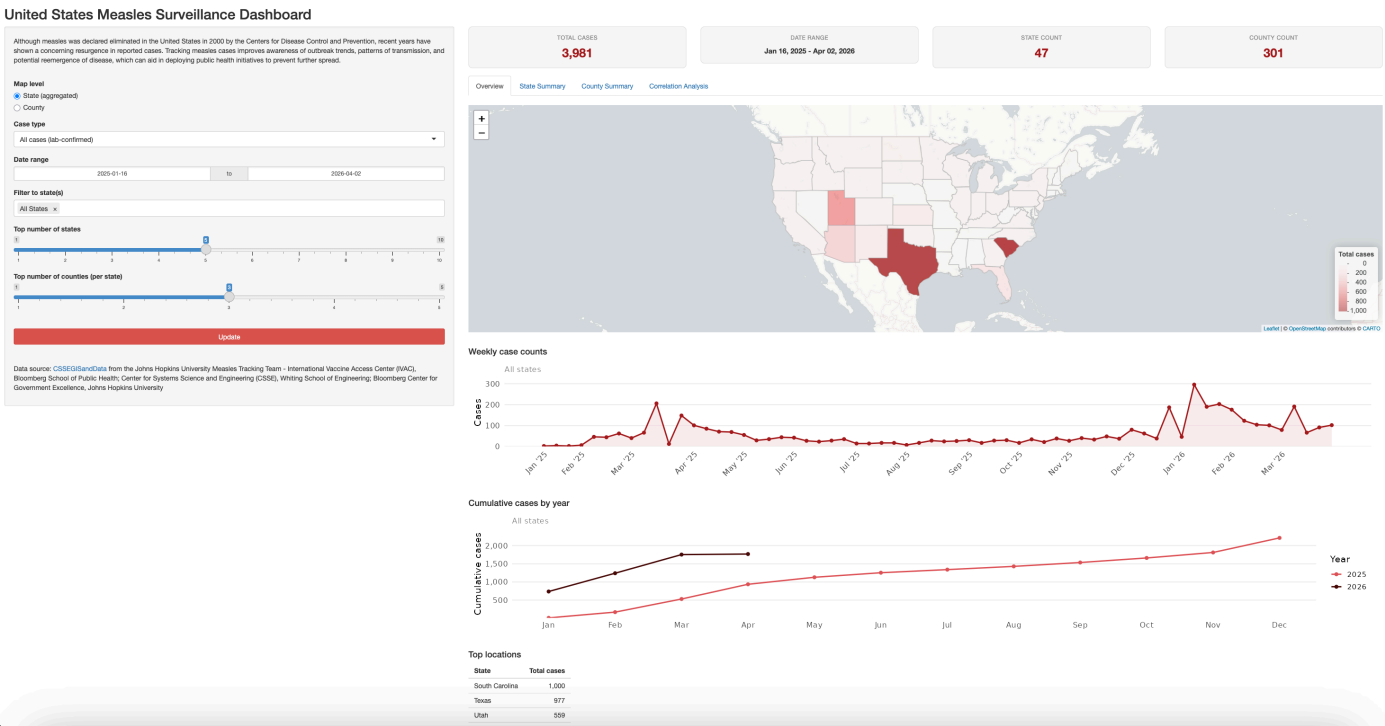
Next

This layout uses a map and time course plots and can be viewed [here](#).



The final Shiny app layout is shown below:

The final layout combines features from initial concepts and can be viewed [here](#).



Dataset Description

The dataset was sourced from https://github.com/CSSEGISandData/measles_data, a GitHub repository that reports data collated by the Johns Hopkins University Measles Tracking Team, which includes members from the International Vaccine Access Center, the Center for Systems Science and Engineering, and Bloomberg Center for Government Excellence. While the repository is regularly updated, our data encompasses the cases that were tracked from January 1, 2025 to April 2, 2026 from commit 2b59391dc10d791051261f5ea8ebbece234ed432. Within this repository, we selected the file `measles_county_all_updates_detailed.csv`, which includes the location of the case (in the format County, State), the Federal Information Processing Standards (FIPS) code for that location, the date of the case (in the format YYYY-MM-DD), the outcome case type (including imported, local, unvaccinated, vaccinated cases), and the value (the count of the confirmed cases that fit into all of the above named specifications). Note that some states only have data available for a subset of the case type.

App Description

For the project, an R Shiny app called "United States Measles Surveillance Dashboard" was created. The app contains five main data analysis displays, which are populated with data gated by filters on the sidebar panel. The first main display is for summary metrics, which include the total number of measles cases, the date range, number of states, and number of counties included based on the user-defined filter settings. The remaining displays appear as tabs below the summary metrics, including the Overview tab, States Summary tab, County Summary tab, and Correlations Analysis tab. The inputs, outputs, and server logic for each of the data analysis displays are described below.

Sidebar Panel

The sidebar panel, implemented using the functions `sidebarLayout` and `sidebarPanel`, contain settings that the user can specify to filter the data. The first one is the map-level selection using `radioButton` for either state or county. If state is selected, state-level geographical boundaries are shown on the map, and the measles cases count is per state. If county is selected, county-level boundaries are shown, and the measles case count is per county. The next filter setting is a single-selection dropdown menu for the measles case type using `selectInput`. Following that is a calendar with `dateRangeInput` where the user can select the start and end dates for the analyses. Below that is a multi-selection dropdown menu for states to include using `selectizeInput`. Next, two sliders with `sliderInput` are included, one for the number of top states and another for the number of top counties per state, which determine the data displayed in table summaries. Finally, an `actionButton` called "Update" is used to apply the filter settings and update the data displays.

Filter type	Input variable name
Map-level selection	<code>input\$map_level</code>
Measles case type	<code>input\$outcome</code>
Date range	<code>input\$date_range</code>
State selection	<code>input\$state_filter</code>
Slider	<code>input\$top_n_states</code>
Slider	<code>input\$top_n_counties</code>
"Update" button	<code>input\$update</code>

Main Panel

The main panel is created with the function `mainPanel` and contains all of the data display outputs, including the summary metrics with `fluidRow` and each of the tabs with `tabsetPanel` and `tabPanel`.

Reactives

The reactives in the analyses are defined here to provide context for the in-depth descriptions of each tab in the subsequent sections. Three types of reactives are used in the analyses: helper function reactives, data filter reactives, and data reactives. The helper function reactives are general support functions whose inputs are dynamically updated variables based on the filter settings. The data filter reactives are used as a preliminary filter on the raw data using the filter settings. The data reactives apply additional filters and data wrangling required for a specific output or data display. The table below outlines all the reactives defined in the R script. In the table, note that `Input` may include input variables from the UI or other reactives.

Reactive	Type	Input	Description	Usage
<code>sel_states</code>	helper	<code>input\$state_filter</code>	returns vector of selected state names with option <code>All States</code> removed, otherwise returns <code>All States</code>	any situation where list of states is needed to filter raw data
<code>plot_subtitle</code>	helper	<code>sel_states</code>	builds subtitle string for overview figures. returns <code>All states</code> if no states selected; returns names if 1-3 states selected; returns count of states if > 3 states selected	Overview tab - weekly case count figure, cumulative cases figure
<code>filtered_raw</code>	data filter	<code>input\$date_range</code> , <code>input\$outcome</code> , <code>sel_states</code> , <code>input\$update</code> . uses helper function <code>filter_by_date_state</code>	applies the date range, outcome type, and state filters to raw dataset, and returns a filtered data frame; gated to <code>input\$update</code> via <code>bindEvent</code>	serves as the preliminary data filter for all downstream summary reactives

Reactive	Type	Input	Description	Usage
<code>filtered_state</code>	data filter	<code>filtered_raw, input\$update</code>	group filtered data by state; gated to <code>input\$update</code> via <code>bindEvent</code>	state-level map, the state top locations table, and the state count summary box
<code>filtered_county</code>	data filter	<code>filtered_raw, input\$update</code>	group filtered data by county; gated to <code>input\$update</code> via <code>bindEvent</code>	county-level map, the county top locations table, and the county count summary box
<code>weekly_cases</code>	data	<code>filtered_raw(), input\$update</code>	group by week	Overview tab - weekly case counts figure
<code>cumulative_cases</code>	data	<code>filtered_raw(), input\$update</code>	group by year and month	Overview tab - cumulative cases by year figure
<code>state_geo</code>	data	<code>sel_states, filtered_state()</code>	downloads state boundary shapefiles and is joined count of selected states	Overview tab - Map
<code>county_geo</code>	data	<code>sel_states(), filtered_county()</code>	downloads county boundary shapefiles and is joined to county count of selected states	Overview tab - Map
<code>state_summary_data</code>	data	<code>filtered_raw(), input\$date_range, sel_states(), input\$top_n_states, input\$update</code>	gated to <code>input\$update</code> via <code>bindEvent</code>	County Summary tab
<code>county_summary_data</code>	data	<code>filtered_raw(), input\$date_range, sel_states(), input\$top_n_states, input\$top_n_counties, input\$update</code>	gated to <code>input\$update</code> via <code>bindEvent</code>	County Summary tab
<code>corr_data</code>		<code>input\$date_range, sel_states(), input\$update</code> . uses helper function <code>filter_by_date_state</code> and <code>ensure_outcome_cols</code>	gated to <code>input\$update</code> via <code>bindEvent</code>	Correlation Analysis tab

Summary Metrics

The summary metrics panel consists of informational boxes with the following information: total number of measles cases, the date range, number of states, and number of counties. This information is updated based on the user-defined filter settings.

The table below summarizes the input variables or reactives, output variables, and functions used on the sever to render the outputs.

Display	Type	Sidebar Filters	Input	Output	Server
Total cases	box	case type, date range, states, update button	sum of <code>filtered_raw()\$value</code> , where <code>value</code> is case count	<code>output\$box_total</code>	<code>renderText</code>
Date range	box	date range, update button	<code>input\$date_range, input\$update</code>	<code>output\$box_dates</code>	<code>renderText, bindUpdate</code>
State count	box	case type, date range, states, update button	<code>n_distinct(filtered_state()\$state)</code>	<code>output\$box_states</code>	<code>renderText</code>
County count	box	case type, date range, states, update button	<code>filtered_county</code> filtered to the number of counties with nonzero cases	<code>output\$box_counties</code>	<code>renderText</code>

Overview

The Overview tab contains data displays that allows the user to explore the dataset at a high level to get a sense of the measles caseload burden geographically and across time. The interactive map summarizes the total case counts for each state or county. The color scale reflects the number of case counts, where the shade of red darkens based on the case count value. In the filter settings, when the map level `state` is selected, the state boundaries are shown with total case counts computed per state. When the map level `county` is selected, the county boundaries are shown with total case counts computed per county. (Note that toggling between `state` or `county` for the map level automatically updates the map boundaries without

using the "Update" button so the user can view both displays with ease. This toggle does not impact the data filtering or wrangling on the backend for other data views.) When users hover over the state or county, the label is shown (either **state** or **county, state**) along with total case count. If specific states are selected in the filter settings, the case count color coding is applied, and all other non-selected states are grayed out. Only data from the selected case type and date range are shown.

Note that a known bug is that there could be minor mismatches between the the county label used in the dataset and the **tigris** package. For example, the dataset lists Upstate as a county in South Carolina. However, Upstate is a descriptor for multiple counties in northern South Carolina. As such, the **tigris** package codes the counties separately, as there is no actual county named Upstate in South Carolina. The result is that case counts for the individual counties are 0, which is technically inaccurate.

Below the map are two figures that summarize measles case loads over time. The data included in these figures is based on the filter settings, specifically the states, case type, and date range. The figure subtitles display information about the states selected. The full state names are listed for up to three states. If more than three states are selected, the number of states is shown. Alternatively, the text **All states** is displayed if that is the option selected.

The first figure below the map is the weekly case counts, with weeks on the x-axis and total cases on the y-axis. Note that the x-axis labels are per month for ease of display. This figure allows the user to visualize when cases begin to increase over time and identify specific weeks when case loads peak. The second figure is the cumulative cases by year, with months on the x-axis and cumulative cases on the y-axis. Years are color coded according to the legend. With this figure, the user can compare caseload trends between different years, in particular for understanding whether the caseload in the current year is improving or worsening compared to previous years. In a real-world situation, the effectiveness of public health initiatives such as vaccination clinics can be generally assessed with visualizations like these to see if caseloads drop over time in specific locations.

The top locations table lists the locations with the most case counts in descending order. When the map level **state** is selected, top states are shown in the table. When the map level **county** is selected, top counties are shown in the table. The **top number of states** slider in the filter settings determines the number of rows in the table.

The table below summarizes the input variables or reactives, output variables, and functions used on the sever to render the outputs.

Display	Type	Sidebar Filters	Input	Output	Server	Helper functions
Map	figure	map level, case type, date range, states, update button	<code>input\$map_level,</code> <code>state_geo()</code> or <code>county_geo()</code> , <code>input\$update</code>	<code>output\$map</code>	<code>renderLeaflet,</code> <code>observe</code> block, <code>bindEvent</code>	<code>make_pal</code>
Weekly case counts	figure	case type, date range, states, update button	<code>weekly_cases,</code> <code>plot_subtitle()</code>	<code>output\$line_plot</code>	<code>renderPlot</code>	<code>plot_subtitle</code>
Cumulative cases by year	figure	case type, date range, states, update button	<code>cumulative_cases,</code> <code>plot_subtitle()</code>	<code>output\$cumulative_plot</code>	<code>renderPlot</code>	<code>plot_subtitle,</code> <code>year_colors</code>
Top locations	table	map level, case type, date range, states, top states slider, update button	<code>input\$map_level,</code> <code>input\$top_n_states,</code> <code>filtered_state()</code> or <code>filtered_county()</code>	<code>output\$top_table</code>	<code>renderTable</code>	

States Summary

The States Summary tab contains an interactive table of summary statistics of the data selected in the analysis at the state level. Data can be sorted in ascending or descending order by clicking on the column title. The data shown is based on the filter settings for case type, date range, and selected states. The **top number of states** slider in the filter settings determines the number of rows in the table, with a maximum of 10.

The columns in the table include the following: state, total count for cases, range, mean, standard deviation, median, interquartile range, proportion of imported cases, proportion of local cases, incidence rate ratio, and number of affected counties.

The descriptive statistics columns (total count, range, mean, standard deviation, median, interquartile range, number of affected counties) are based on the case type selected. The statistics are calculated over counties within a given state.

The proportion and rate columns (proportion of imported cases, proportion of local cases, incidence rate ratio) are based on all case types. The proportion of imported cases is calculated as the number of imported cases divided by the sum of imported and local cases. The proportion of local cases is calculated as the number of imported cases divided by the sum of imported and local cases. The incidence rate ratio (IRR) is calculated as the ratio of unvaccinated cases to vaccinated cases.

- IRR > 1: indicates that measles incidence is higher in the unvaccinated group than in the vaccinated group; as such, the vaccine may reduce the risk of disease
- IRR = 1: indicates that measles incidence is the same in both vaccinated and unvaccinated groups; as such, this may be an initial warning of waning vaccine immunity
- IRR < 1: indicates that measles incidence is lower in the unvaccinated group than in the vaccinated group; as such, this may be due to severe waning vaccine immunity

The table below summarizes the input variables or reactives, output variables, and functions used on the sever to render the outputs.

Display	Type	Sidebar Filters	Input	Output	Server	Helper functions
State summary	table	case type, date range, states, top states slider, update button	<code>state_summary_data()</code>	<code>output\$state_summary_table</code>	<code>renderDataTable</code>	<code>build_outcome_wide</code> (<code>filter_by_date_state</code> , <code>ensure_outcome_col</code> , <code>add_proportion_cols</code>)

County Summary

The County Summary tab contains an interactive table of summary statistics of the data selected in the analysis at the county level. Data can be sorted in ascending or descending order by clicking on the column title. The data shown is based on the filter settings for case type, date range, and selected states. Both the `top number of states` slider and `top number of counties (per state)` slider in the filter settings determine the number of rows in the table. For example, with the default settings, `top number of states` is 5 and `top number of counties (per state)` is 3. This means that for the top 5 states, data for the top 3 counties in each of those states is shown. The same columns are used as in the State Summary, with the addition of a column for county. Note that at the county level, the statistics and other metrics are calculated over reporting days for a given county.

The table below summarizes the input variables or reactives, output variables, and functions used on the sever to render the outputs.

Display	Type	Sidebar Filters	Input	Output	Server	Helper functions
County summary	table	case type, date range, states, top states slider, top counties slider, update button	<code>county_summary_data()</code>	<code>output\$county_summary_table</code>	<code>renderDataTable</code>	<code>build_outcome_wide</code> (<code>filter_by_date_state</code> , <code>ensure_outcome_col</code> , <code>add_proportion_cols</code>)

Correlation Analysis

The Correlation Analysis tab is for performing exploratory correlations between predefined case types in the dataset to address two scientific questions. The data filters applied are the date range and selected states. Note that case types are fixed by the correlation being examined, thus the case type filter is not used.

The first inquiry is understanding the relationship between imported cases and local transmission. Imported cases are known cases where measles was contracted outside of the given location. Whether increasing imported cases lead to higher local transmission is investigated. The second inquiry is understanding the relationship between unvaccinated cases and local transmission. Whether increasing unvaccinated individuals with measles lead to higher local transmission is examined. If this information is tracked, public health initiatives can be deployed to locations with high imported or unvaccinated cases to block the spread of local transmission.

Figure 1 is a scatter plot for imported cases versus local transmission cases. Figure 2 is a scatter plot for unvaccinated cases versus local transmission cases. In both figures, a linear trend line is shown with shading as the 95% confidence interval. The Spearman correlation was performed, since it does not require the assumption of normality and is less sensitive to outliers. The statistics summary tables below each figure list the correlation value (ρ), p-value, and sample size. The analyses uses data at the county level based on the selected states.

The table below summarizes the input variables or reactives, output variables, and functions used on the sever to render the outputs.

Display	Type	Sidebar Filters	Input	Output	Server	Helper functions
Fig 1 (imported vs local cases)	figure	date range, states, update button	<code>corr_data()</code>	<code>output\$corr_plot_imp_local</code>	<code>renderPlot</code>	<code>build_corr_plot</code>
Stats summary for Fig 1	table	date range, states, update button	<code>corr_data()</code>	<code>output\$corr_table_imp_local</code>	<code>renderTable</code>	<code>build_corr_table</code>
Fig 2 (unvaccinated vs local cases)	figure	date range, states, update button	<code>corr_data()</code>	<code>output\$corr_plot_unvacc_local</code>	<code>renderPlot</code>	<code>build_corr_plot</code>
Stats summary for Fig 2	table	date range, states, update button	<code>corr_data()</code>	<code>output\$corr_table_unvacc_local</code>	<code>renderTable</code>	<code>build_corr_table</code>

Analyses and Insights

For this sample analyses, the default filter settings are used. This includes the case type as **All cases (lab-confirmed)**, all dates (**2025-01-16 to 2026-04-02**), all states, maximum of 5 for **top number of states**, and maximum of 3 for **top number of counties (per state)**.

On the summary metrics upper panel, we see a total number of 3,981 cases from 47 states and 301 counties within the United States. The Overview tab describes the geographical trends and time course of the measles caseload. At the state level, it is evident that Texas and South Carolina visually stand out as states with the highest number of cases. At the county level, Gaines, Texas, and Mohave, Arizona are the counties with the highest number of cases. (Note the known data display bug as described in the Overview section above, regarding data labeling issues that may impact the display of county-level data on the map.) In the weekly case counts figure, the caseload peaks are sometime during March 2025 with more than 200 cases and January 2026 with more than 300 cases. Generally, caseloads are higher than baseline from February-May 2025 to January-April 2026. In the cumulative cases by year figure, the measles caseload is higher overall as compared to last year within the January to April timeframe. The top locations table confirms what was observed on the map views. South Carolina, Texas, Utah, Arizona, and Florida have the highest cases nationally. In addition, these counties have the highest cases: Upstate, South Carolina, Gaines, Texas, Mohave, Arizona, Southwest Health District, Utah, and Hudspeth, Texas.

In the State Summary tab, the top 5 states are listed in a table in descending order of case count. A few interesting observations can be made. South Carolina has a very high incidence rate ratio (IRR), meaning that unvaccinated cases are higher than vaccinated cases. That certainly aligns with the prior observation that South Carolina is one of the states with the highest number of cases, and contains a county with one of the highest caseloads. In addition, Texas has the highest number of affected counties, which may speak to greater within-state disease spread. Another noteworthy metric is that Florida has the highest proportion of imported cases.

In the County Summary tab, the top 3 counties from each of the top 5 states are listed in a table in descending order of case count. Similar observations can be made as the State Summary in terms of IRR. With the county-level data, it is clear that the high cases in Upstate, South Carolina and Southwest Health District, UT contribute to that. Another observation is that Hudspeth, Texas has the highest mean and median cases, meaning it has the most cases reported per day.

The Correlation Analysis tab shows results from Spearman rank correlations that were performed. Spearman correlations are used since it does not require the assumption of normality and is less sensitive to outliers. All available county-level data for the specified case types are included. For the imported cases versus local transmission correlation, a negative correlation (-0.366) with $p < 0.001$ was found. One interpretation is that herd immunity is effective at blocking the spread of measles, since imported cases enter a location whose population has a high vaccination coverage, leaving fewer susceptible people to infect. For the unvaccinated cases versus local transmission correlation, a weak positive correlation (0.126) with $p = 0.029$ was found. This may suggest that gaps in vaccination may be driving the local spread of measles. One of the challenges with this analyses is that data for a given case type are not available for some of the states and/or counties, which may skew analyses at the national level.

GitHub and Shiny App Deployment

The code for the Shiny app can be found on the team's public GitHub repository [here](#).

The app is hosted online at shinyapps.io [here](#).

References

U.S. Measles Cases Hit Highest Level Since Declared Eliminated in 2000 | International Vaccine Access Center. (2025, July 7). International Vaccine Access Center. <https://publichealth.jhu.edu/ivac/2025/us-measles-cases-hit-highest-level-since-declared-eliminated-in-2000>

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